

LAYER 4 · SYSTEM GOVERNANCE & ASSURANCE TIER [0.1–1 Hz]

Policy Supervision & Safety Cases

Ch 14 Shared Autonomy

Policy Blending · Takeover Latency

Ch 15 Verification

Sim → PIL → HIL → In-Situ Ladder

Ch 16 Safety Cases

Goal Structuring · UL 4600 / ISO

Ch 17 Frontier Limits

Observational Indistinguishability

LAYER 3 · THE BRAIN: COGNITIVE DELIBERATION TIER [1–50 Hz · Linux MPU / Edge NPU]

High-Capacity Foundation Models, Spatial Perception, Latent World Belief, and Action Chunking (Ch 03 Brain)

1. Ingestion (Ch 05)

MIPI CSI-2 · Ring Buffers

2. Perception (Ch 08)

$o_t \rightarrow z_t$ · ViT Encoders

3. Memory (Ch 09)

$z_t \rightarrow b_t$ · SE(3) Belief

4. Intent (Ch 10)

$b_t \rightarrow L_t$ · Expiring Lease

5. Planner (Ch 11)

$L_t \rightarrow \hat{u}$ · Diffusion / ACT

PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (Shared SRAM Mailbox · Untrusted $\hat{u}_{t:t+H} \rightarrow$ Verified u^*)

Proposal $\hat{u}$

LAYER 2 · THE NERVOUS SYSTEM: REAL-TIME SAFETY & TIMING TIER [1000 Hz · Bare-Metal MCU]

Zero-Allocation Deterministic Silicon, Hardware Watchdogs, and Safety Refusal (Ch 04 Nervous System)

1 kHz Reflex Enforcer (Ch 12)

Control Barrier Filter $h(\mathbf{x}) \geq 0$ · Vetoes Unsafe $\hat{u}$

Silicon Placement & Bus QoS (Ch 13)

Interconnect QoS · SRAM Seqlock · Timers

Certified Actuator Latch

Latches u^* into PWM Compare Registers

Certified u^*

LAYER 1 · THE PHYSICAL BODY & CONTINUOUS PLANT [Continuous Dynamical Mechanics]

Matter, Inertia, Transduction, Friction, and Thermal Limits (Active: Ch 01 Boundary · Ch 02 Body · Ch 06/07 Reality Gap)

Power Inverters & Gate Drivers

PWM Switching → Phase Current I_{phase}

Mechanics, Torque & Momentum

Lorentz Force → Torque τ → Momentum $p = mv$

Causal Boundary & Dissipation

Stopping Envelope $d_{\text{stop}} \leq d_{\text{clear}}$ · Joule Heat

Endogenous Sensory Shift ($o_{t:t+1}$)